

Machine Learning Applied to Time Series

What Can ML Do with Time Series Data?

1. Forecasting / Prediction

1. Predict **future values** based on past trends.
2. **Example:** Predict Japan's future birth & death rates to help government plan policies.

2. Back casting / Historical Projection

1. **Project into the past** to estimate values before actual data was collected.
2. Helps understand how current trends evolved.

3. Imputation (Filling Missing Data)

1. Fill **gaps in time series** where no data exists.
2. **Example:** Missing years in Moore's Law chart because no chips were released.

4. Anomaly Detection

1. Identify unusual patterns or spikes.
2. **Example:** Detect sudden traffic spikes in website logs → possible DDoS attack.

5. Pattern Recognition

1. Recognize **repeating structures** in time series to identify what generated it.
2. **Example:** Speech recognition analyzing sound wave time series to detect words/subwords.

➤ Key takeaway:

Machine learning for time series isn't just about predicting the future — it can also fill in the past, repair missing data, detect unusual events, and recognize underlying patterns.
